A Scalable Recurrent Structure With Fast Transfer Learning for Lithium-Ion Battery State of Charge Estimation at Different Ambient Temperatures

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***Abstract*— State-of-charge (SOC) is a critical parameter of battery management systems to ensure safe, efficient, reliable and durable battery operations. However, SOC cannot be directly measured and is highly sensitive to different temperatures. Thus, the SOC estimation accuracy and uncertainty management are significant for robust control and energy dispatch. Gaussian process regression (GPR) is thus becoming appealing due to its non-parametric and interpretable probabilistic advantages. But the scalability of GPR is still challenging, suffering from cubic complexity. To solve these challenges, a SOC estimation method is proposed by an end-to-end scalable deep recurrent structure with fast transfer learning at different temperatures. First, convolutional and recurrent neural networks are employed to catch nonlinear temporal dependency within measurements. Second, a GPR layer is concatenated after neural networks, so that the estimation can be quantified with uncertainty while retaining nonlinear expressive ability. Then, a non-parametric fast transfer learning is designed to realize fast transfer between different temperatures. Next, structured sparse approximations and a semi-stochastic gradient procedure are established for scalable training. Finally, the accuracy and fast transfer of the proposed structure are verified through comparison. The structure demonstrates the state-of-the-art performance on estimation accuracy and efficiency with transfer learning faster than fine-tuning strategy by two orders of magnitude.**

***Index Terms*— Lithium-ion batteries (LiBs), data-driven, state of charge (SOC) estimation, transfer learning, different ambient temperature.**

I. Introduction

WITH the rising concerns about energy crisis and global warming, electrification of transport sector has emerged as one of the greatest challenges of the latest decades. Electric vehicles (EVs) have evoked tremendous interest by both industry and academia [1]. Rechargeable lithium-ion batteries (LIBs) are currently the best candidate for EVs due to their desirable merits of reasonable energy/power density, cycle life, and low self-discharge rate [2]. While the increasing energy/power density can promote the adoption of EVs, it can also lead to potential efficiency and safety risks. Another critical barrier to widespread adoption of EVs is range anxiety. An advanced battery management system (BMS) that can provide running ability information about monitored batteries is thus essential to accurately determine the state of charge (SOC) and optimize battery behaviors.

In essence, SOC is an equivalent of a fuel gauge for EVs, defined as the ratio between the remaining capacity and total capacity. It is an essential parameter for diverse BMS facilities, such as estimating driving range, health diagnosis [3], optimal charging control [4], [5], balancing [6], [7], etc.. The SOC cannot be directly measured. Instead, it must be estimated by measurable quantities, such as terminal voltage, current, and temperature provided by BMS. In this sense, the estimation accuracy and uncertainty quantification are significant, which enables BMS to make robust control, optimization, and avoid unexpected failures and losses [8]. The absence of uncertainty quantification may lead to unreliable predicted results without confidence. [9]. The uncertainty quantification contributes to a well-designed maintenance strategy by providing battery intrinsic state variation and reporting uncertainty level of estimated values [10]. However, accurate and robust SOC estimation is still a challenging task due to highly nonlinear dynamic behaviors, variations of temperatures and loading profiles, and complicated degradation trajectories.

To date, a variety of SOC estimation technologies have been developed, which can be categorized into three main classes, including open-loop calculation, model-based, and data-driven approaches.

1) Open-Loop Calculation: Coulomb-counting and open-circuit voltage (OCV)-SOC table look-up methods are two typical open-loop calculation methods. The Coulomb-counting method, also called Ampere-hour integration method, directly calculates SOC using an integration of measured current. It has predominated in developed BMS owing to the ease of implementation and simplicity [11]. However, due to a lack of feedback calibration, the initial errors and cumulative noises of current measurements would lead to incorrect or even divergent SOC estimation. The OCV-SOC table look-up

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method employs the stable OCV-SOC relationship of LIBs. Once OCV is correctly measured, the SOC can be directly calculated using the offline calibrated OCV-SOC table. The stability justification of the OCV-SOC relationship can be found in [12]. However, OCV can only be measured when a battery reaches its equilibrium status, which thus limits the feasibility of online and real-time implementation. It is worth noting that these open-loop calculation methods use independent current or voltage information and thus show obvious limitations. Once certain conditions or assumptions are not satisfied in laboratory tests, these methods are less accurate than expected and not suitable for practical applications.

2) Model-Based: To overcome the limitations of open-loop calculation methodologies, the model-based methods adopt closed-loop observer designs to online estimate battery SOC. These methods first establish a battery model that takes battery SOC as a hidden state variable. Then, an adaptive observer is designed to fulfill state estimation. The research community has developed diverse technologies to solve problems in terms of linearization accuracy [13], robustness against parameter disturbance [14], [15], adaptability to sensors' noises [16], model adaptability to parameter changes [14], [17], and convergence speed [18], etc.. Despite these efforts in observer algorithms, the accuracy, adaptability, and robustness of battery models have remained a critical technological bottleneck in these model-based methods. Currently, the most used methods include the equivalent-circuit model (ECM) [3], [14] and the physics-based model (PBM) [13], [19]. The former is essentially derived from empirical knowledge and experimental data, while the latter starts from the first-principles characterization of electrochemical processes including thermodynamics, reaction kinetics, and mass transport. In ECMs, the batteries are represented by groups of electrical components such as nonlinear sources, resistors, capacitors, and even fractional-order elements. The advantages of ECMs are their high computational efficiency and simplicity. However, they generally present limited accuracy within a very narrow operating condition. The reason is that the parameters of ECMs preserve no physical meaning, and a parameterization process is necessary to calibrate the model fitness. In addition, the accuracy and robustness will decline inevitably due to parameter variations caused by temperature changes and battery degradation. On the contrary, PBMs show more accurate behavior descriptions than ECMs. The bottleneck of PBMs is their high computational complexity since they are generally described by a set of partial differential equations, which require intensive computations to solve. Thus, it is impractical to embed PBMs into an onboard BMS controller for real-time applications [20]. Despite the complications during developing more precise and efficient models, unpredictable disturbances, model mismatches, and identification issues pose obstacles for the model-based approaches.

3) Data-Driven: With the advancement of machine-learning algorithms and intelligent computation technologies, data-driven machine-learning methods are becoming more and more promising for intelligent BMS design. There are three time-dependent data streams describing a battery in operation, i.e., current, voltage, and temperature. The role of any data-driven methods is to map the time-evolution of these variables to battery SOC [21]. These methods typically include two major steps: offline training and online prediction. During the training process, the parameters of a learner or fitting function are optimized using experimental training data. Then, the trained learner predicts for other battery systems. As reviewed in [20], the most mentioned learners include support vector machine (SVM) [22], artificial neural networks (ANN) [23], and Gaussian process regression (GPR) [24]. There are also many variations using the latest machine learning technologies, such as using deep neural network (DNN) to improve accuracy [25], transfer learning to adapt different distributions of train and test data, and transfer learning for different chemical characteristics [26], etc. Here, this study highlight the gated recurrent unit (GRU) structure [27] and GPR for the convenience of dealing with sequential data and quantifying estimate uncertainties, respectively. The former provides hidden states that store the long-term dependencies of battery dynamics, which is thus suitable for learning nonlinear behaviors of sequential data [28]. Besides, convolutional neural network (CNN) [29], [30] is often selected as a generic component for deep neural structure to extract local hidden information and establish an end-to-end framework [31]. However, the important question of modeling and propagating uncertainty across time has rarely been addressed by ANN-based models [32]. The latter is a stochastic method that delivers a probability distribution of possible predictions and thus allowing a principled representation of uncertainty and non-parametric flexibility. Additionally, The limitation of GPR lie in two aspects. First, the regular GPR is not suitable to model sequential behaviors, and thus recurrent GPR (R-GPR) that the SOC estimate in the previous discrete-time instant is fed back to the input [33]. The RGPR mimics the standard recurrent neural network (RNN) by substituting every layer with a Gaussian process (GP). The RGPR seems problematic in terms of its implementation and scalability to massive datasets. In addition, RGPR only captures short-term battery dynamics, while long-term behaviors SOC trend might be absent. Thus, it is meaningful to combine GRU structural and GPR for battery SOC estimation with the purpose of achieving the benefits and avoiding shortcomings of both techniques. A relevant work is [8], which combined long short-term memory (LSTM) networks and GPR for state-of-health estimation. This work trained the LSTM and GPR separately, one for long-term dependence and the other for uncertainty quantification. Its accuracy is, however, mainly determined by LSTM. The mechanical superposition of the two techniques seems not to outperform the two methods essentially. Moreover, achieving compatibility between the neural networks and Bayesian methods like GPR is non-trivial, due to fundamental distinctions in their architectures, training paradigms and parameter optimization strategies. GRU is a simpler architecture than LSTM with similar ability to deal with long-term sequential data. Thus, SOC estimation based on more organic unity of GRU and GPR is drawing more attention.

LiBs work under different ambient temperatures in real-world scenarios and are highly sensitive to temperature changes. However, training data for SOC estimation model is always collected at one or several fixing ambient temperatures. The domain shift between target test battery data and source training data can cause severe prediction errors [34]. Transfer learning has been proposed for its power to boost accuracy with deficient data in many battery management algorithms [35]. Transfer learning has also been introduced to SOC estimation to adapt to different ambient temperatures [36]. In the existing literature, transfer-based SOC estimation methods can be mainly divided into the model parameter domain adaptation strategy and the fine-tuning strategy [37]. The domain adaption methods reduce the discrepancy of feature distribution between source and target domains by introducing the maximum mean discrepancy (MMD) loss [38]. On the other hand, the fine-tuning strategy utilizes the source domain information by treating parameters from the model pre-trained in the source domain as initial values for target batteries [26]. The fine-tuning strategy is easy to implement and provides fair performance but requires extra retraining time which, limits its practical application. Therefore, fast transfer learning based on fine-tuning is worth investigating.

The main research gaps in the field of LIB SOC estimation are summarized into three aspects from the preceding discussion. First, SOC estimation at different temperatures still lacks an accurate and universal solution since LiBs are highly sensitive to temperature changes though many estimation methods have been proposed. Second, the SOC estimation results may be unreliable when performed under unknown operating conditions without confidence interval (CI). Lastly, a number of studies applied transfer learning at different temperatures to estimate SOC, but the conventional parametric fine-tuning strategy leads to more time consumption, which limits their feasibility.

To bridge these gaps, this article proposes an end-to-end SOC estimation method based on a deep recurrent structure with fast transfer learning at different temperatures, considering the sequential essence of battery SOC signal, uncertainty quantification requirements, and scalability to massive datasets of different batteries under different loading conditions. Specifically, several main contributions are summarized as follows.

1) This study proposed a scalable recurrent structure with fast transfer learning through compatible integration of deep neural networks and Bayesian methods in a modified implementation. The proposed deep recurrent structure can fully encapsulate the advantages of GRUs for training recurrent structures while allowing for a principled quantification of uncertainty. Thus, it can not only learn long-term dependence of battery dynamics but also represent and quantify predictive uncertainty in a probabilistic Bayesian manner.
2) A novel fast transfer learning is designed by optimizing the hyperparameters of the GPR kernels. This means that the heavy computational procedure of retraining neural network weights is bypassed. The fast transfer learning fashion makes it possible to reduce time consumption by orders of magnitude while retaining satisfactory accuracy.
3) A provably convergent semi-stochastic optimization algorithm is introduced to jointly train the RNN and the GPR layers, which improves scalability of SOC estimator for large-scale datasets. This advancement enables effective utilization of massive battery test data across diverse operational scenarios.

II. Methodology

In this section, the proposed pipeline for battery SOC estimation and transfer learning at different ambient temperatures is given first. Then, the related technologies, namely, CNN for recognizing the latent features from raw data, RNN for learning sequential data prediction, GRU module for capturing long-term dependence, and GPR for quantifying uncertainty will be introduced respectively. Next, the proposed end-to-end scalable deep recurrent structure, the training algorithm, as well as the implementation details will be introduced. Finally, the fast transfer learning at different ambient temperatures will be presented.

A. Pipeline for SOC Estimation and Transfer Learning

To accurately and robustly estimate battery SOC and fast transfer learning for different ambient temperatures, a scalable SOC estimation and transfer learning pipeline is proposed, as shown in Fig. 1, which incorporates three hierarchical steps, namely, (a) source domain data or target domain data curation, (b) offline training for the neural networks, (c) online prediction and fast transfer learning. From a fine-tuning strategy perspective, the CNN and the GRU neural networks are frozen after offline training by source domain. Conversely, the hyperparameters of GPR layers will be further optimized with input data in the target domain.

First, to increase the local information perception ability, the battery time-dependency measurable data streams, i.e., V, I, and T are reshaped into a two-dimensional array as the input of CNN. In this way, the neighborhood dependencies and time-domain locality relationship between these variables can be extracted by CNN. The two dimensions consist of type dimension and sampling dimension. The type dimension includes the data type, i.e., V, I, and T. And the sampling dimension stores the specific values of each data type at 1 Hz sampling frequency. For each prediction point, a slice from the two-dimensional array will be used for online prediction, offline training, or fast transfer learning.

Second, a scalable deep recurrent structure is designed to map the relationship between the selected slice and battery SOC. The designed deep structure concatenates CNN, GRU, and GPR layers in series, which helps to take advantage of CNN in extracting local latent relationships, GRU in capturing the long-term dependencies of sequential data, and GPR in non-parametric uncertainty quantifying.

Third, the proposed deep structure enables fast transfer learning owing to the non-parametric advantages. When a new target domain data arrives, the neural network part of the pre-trained structure does not need to be retrained and bypass

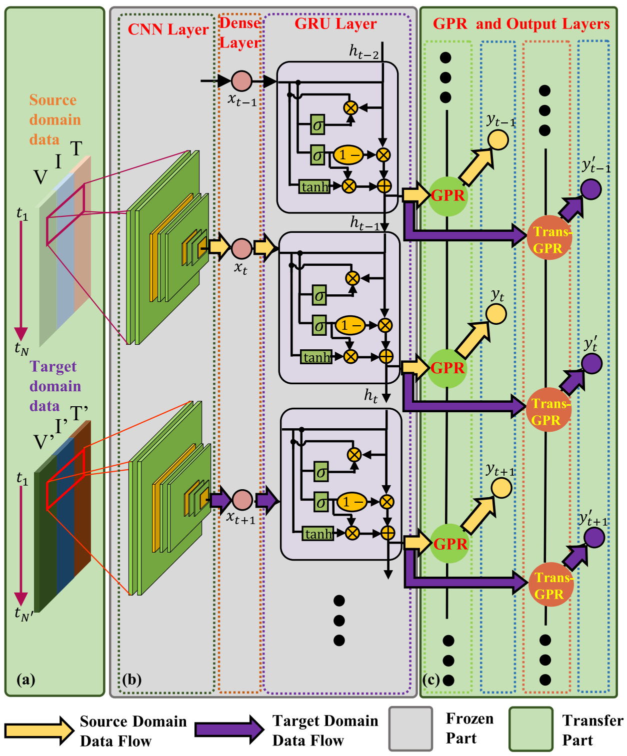

Fig. 1. The proposed scalable recurrent SOC estimation framework with fast transfer learning.

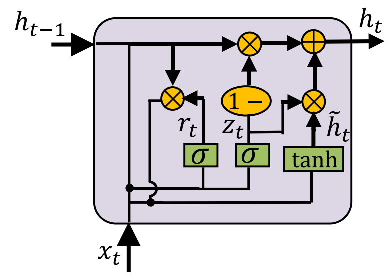

Fig. 2. The gated recurrent unit structure.

a plethora of network weights to be updated. Instead, only a few hyperparameters of GPR layers will be optimized to obtain satisfactory prediction accuracy.

As shown in Fig. 1, the proposed method can establish a nonlinear map between the measured data streams and estimated SOC with uncertainty quantifying. It is worth noting that the fast transfer learning technique makes online transfer training with target domain data possible.

B. The Proposed Scalable Deep Recurrent Structure

To fully encapsulate the local perception ability and long-term dependency characteristics of the battery operational data streams, while retaining probabilistic expression and non-parametric transfer learning advantage, an end-to-end scalable deep recurrent structure with fast transfer learning is developed based on properties of CNN, long-term recurrent structure, GPR, and transfer learning, which is denoted as CL-GPT. The designed structure is illustrated by part (b) and (c) in Fig. 1.

1) Convolutional Neural Network: To accomplish the end-to-end estimation, CNN is a significant component due to its advantages of automatic feature extraction and high regression ability and tells many successful stories in perceiving implicit relationships from raw data streams. Two pivotal operations of CNN in the proposed structures are local connection and full connection, two-dimensional convolutional layers and dense layers in detail, respectively. The local connection constructs several convolutional kernels to extract features from the multi-dimensional input data curation. It is worth noting that the convolutional kernels only focus on nearby relations, which resuls in reinforcing local feature extraction and reduction of computational intensity and data storage. The formulation of a two-dimensional convolutional layer is presented as,

$$z_{i,j}^{(l+1)} = f\left(\sum_{p=i-m/2}^{i+m/2} \sum_{q=j-n/2}^{j+n/2} z_{p,q}^{(l)} w_{p,q}^{(l)} + b^{(l)}\right) \tag{1}$$

where $z^{(l)}$ presents the output latent variables of the lth convolutional layers, w^l and $b^{(l)}$ denotes the weights and bias of the convolutional kernel in the lth layer, and m, n is the length and width of convolutional kernel, respectively. f is the nonlinear activation function and the rectified linear unit (ReLU), expressed as $f(x) = \max(0, x)$, is generally selected as the activation function.

The dense layer, similar to regular neural network, connects all the outputs of the previous layer with respective weights and biases to generate each output of this layer. The dense layers strengthen connections between multi-dimensions but are not sufficient to capture long-term temporal dependency. Thus further processing with recurrent structure is required.

2) Gated Recurrent Unit Network: Recurrent structures have been recognized as a paradigm for dealing exploiting sequential data streams. RNNs process the sequence-to-real regression problem, by using linear parametric maps. The sequential characteristics are effectively represented by an entire latent sequence $\{\mathbf{h}_i\}_{i=1}^{t}$. However, the major weakness of RNNs is the vanishing gradient problem during error propagation.

To overcome the vanishing gradients, GRU introduces a gating mechanism and embeds two differentiable gate variables into the hidden neurons of RNNs. As shown in Fig. 2, GRU networks are simpler than LSTM architecture with one fewer gate and have no additional memory cell. The simplicity makes the training of GRU easier and more effective.

The r_t and z_t represent the reset gate and update gate allowing the recurrent network to decide whether to keep or erase previous hidden states. The gating mechanism boosts the ability to capture long-term temporal relationships. However, the GRU can only output a point estimate for battery SOC, which is not practical to account for the statistical characteristics and battery operational planning. In addition, the parametric estimation brings on a retraining burden for transfer learning.

3) Gaussian Process Regression Layer: To describe battery SOC with uncertainty quantifying, GPR, a Bayesian non-parametric model, can be applied to express probabilistic predictive distribution. Given training set with inputs $\mathbf{X}$ and targets $\mathbf{y}$ with the assumption of additive Gaussian noise and a GP prior on function f. The predictive distribution at the test point $\mathbf{x}_*$ is evaluated as:

$$\mathbf{f}_*|\mathbf{x}_*, \mathbf{X}, \mathbf{y} \sim \mathcal{N}(\mathbb{E}[\mathbf{f}_*], \mathrm{Cov}[\mathbf{f}_*]) \tag{2}$$

with

$$\mathbb{E}[\mathbf{f}_*] = \boldsymbol{\mu}_{\mathbf{x}_*} + K_{\mathbf{x}_*,\mathbf{X}}[K_{\mathbf{X},\mathbf{X}} + \sigma^2 I]^{-1}\mathbf{y} \tag{3a}$$

$$\mathrm{Cov}[\mathbf{f}_*] = K_{\mathbf{x}_*,\mathbf{x}_*} - K_{\mathbf{x}_*,\mathbf{X}}[K_{\mathbf{X},\mathbf{X}} + \sigma^2 I]^{-1} K_{\mathbf{X},\mathbf{x}_*} \tag{3b}$$

where $K_{\mathbf{X}_*,\mathbf{X}}$,$K_{\mathbf{X},\mathbf{X}}$,$K_{\mathbf{X}_*,\mathbf{X}_*}$, and $K_{\mathbf{X},\mathbf{X}_*}$ are covariance matrices of kernel function, $\kappa(\cdot,\cdot)$, evaluated at the corresponding points, $\mathbf{X}$ and $\mathbf{x}_*$. $\boldsymbol{\mu}_{\mathbf{X}_*}$ is the mean function evaluated at $\mathbf{x}_*$. GPRs can capably learn the non-parametric functional relationships between inputs and targets. However, GPRs are limited to pairwise relationships within the input data and can not perceive long-term dependencies within temporal sequences. Moreover, GPRs suffer a heavy computational burden with data size becoming large, leading to scalability issues.

4) Deep Recurrent Structure: Motivated by the merits and demerits of CNN, GRU, and GRP, an end-to-end scalable deep recurrent structure is established to encapsulate the local relationship perception of CNN, the series temporal dependency learning of GRU, and the non-parametric advantage of GPR. From the perspective of the probabilistic GPR framework, the proposed model can be interpreted as a Gaussian process with a recurrent kernel, of which the neural networks, i.e., CNN and GRU, play the role. Specifically, the original input space is transformed by the CNN and GRU neural network. This can be interpreted as building a recurrent kernel directly in the transformed space, as described in Fig. 1.

For the probabilistic Bayesian model, it is natural to realize to use the negative log marginal likelihood (NLML) objective loss function to optimize model parameters and hyperparameters, formulated as follows,

$$\mathcal{L}(K) = \mathbf{y}^\top (K_y + \sigma^2 I)^{-1}\mathbf{y} + \log\det(K_y + \sigma^2 I)] \tag{4}$$

where $K_y + \sigma^2 I (\triangleq) K$ is the Gram kernel matrix and implicitly derived from the GPR kernel hyperparameters, $\boldsymbol{\theta}$, and the parameters of the neural network transformation, $\boldsymbol{\phi}(\cdot)$, denoted W. The proposed probabilistic model is optimized by minimizing $\mathcal{L}$ with respect to both $\boldsymbol{\theta}$ and W.

To achieve a conjugate gradient descent fashion, the partial derivatives of the loss function $\mathcal{L}$ w.r.t. $\boldsymbol{\theta}$ and W are needed. Since the GPR layer is at the shallow surface, the derivative of the NLML loss function w.r.t. $\boldsymbol{\theta}$ is standard and holds the following form [39]:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}} = \frac{1}{2}\mathbf{tr}\left(\left[K^{-1}\mathbf{y}^\top\mathbf{y}K^{-1} - K^{-1}\right]\frac{\partial K}{\partial \boldsymbol{\theta}}\right) \tag{5}$$

where $\partial K/\partial\boldsymbol{\theta}$ has an analytic form for corresponding kernel function, $\kappa(\cdot,\cdot)$, at most of the time.

In contrast to surface layers, the deeper neural network layer optimization is nontrivial, and chain rules are employed:

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{1}{2}\mathbf{tr}\left(\left[K^{-1}\mathbf{y}^\top\mathbf{y}K^{-1} - K^{-1}\right]\frac{\partial K}{\partial W}\right) \tag{6}$$

where $\partial K/\partial W = \partial K/\partial H \cdot \partial H/\partial W$, $H \in \mathbb{R}^N$ is the CNN and GRU neural network transformation outputs. As $\partial K/\partial H$ has no straightforward solutions and calls for an element-wise derivative, the i^{th} and j^{th} entry of kernel matrix K_{ij} w.r.t. the m-th parameter of the neural network, H_m, is elaborated as follows:

$$\frac{\partial K_{ij}}{\partial W_m} = \mathbf{tr}\left(\left(\frac{\partial K_{ij}}{\partial H}^\top\right)\frac{\partial H}{\partial W_m}\right). \tag{7}$$

Noting that K_{ij} only depends on the i-th and j-th elements of H for which the kernel is being computed, (7) can be rewritten as:

$$\begin{aligned}\frac{\partial K_{ij}}{\partial W_m} &= \left(\frac{\partial K_{ij}}{\partial \mathbf{h}_i}\right)^\top \frac{\partial \mathbf{h}_i}{\partial W_m} + \left(\frac{\partial K_{ij}}{\partial \mathbf{h}_j}\right)^\top \frac{\partial \mathbf{h}_j}{\partial W_m}\\ &= \left(\frac{\partial K(\mathbf{h}_i,\mathbf{h}_j)}{\partial \mathbf{h}_i}\right)^\top \frac{\partial \mathbf{h}_i}{\partial W_m} + \left(\frac{\partial K(\mathbf{h}_i,\mathbf{h}_j)}{\partial \mathbf{h}_j}\right)^\top \frac{\partial \mathbf{h}_j}{\partial W_m}\end{aligned} \tag{8}$$

Since the derivative $\partial K(\mathbf{h}_i,\mathbf{h}_j)/\partial \mathbf{h}_i$ is feasible, (7) is accessible by plugging (8) into (6) and arrives at the following form:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial W_m} &= \frac{1}{2}\sum_{ij}\left(K^{-1}\mathbf{y}^\top\mathbf{y}K^{-1} - K^{-1}\right)_{ij}\\ &\left\{\left(\frac{\partial K(\mathbf{h}_i,\mathbf{h}_j)}{\partial \mathbf{h}_i}\right)^\top \frac{\partial \mathbf{h}_i}{\partial W_m} + \left(\frac{\partial K(\mathbf{h}_i,\mathbf{h}_j)}{\partial \mathbf{h}_j}\right)^\top \frac{\partial \mathbf{h}_j}{\partial W_m}\right\}\end{aligned} \tag{9}$$

It can be observed from (5) and (9) that both GPR layer hyperparameter and neural layer parameter optimization are involved with the inverse of the kernel matrix, K, which typically suffers the $\mathcal{O}(n^3)$ computational complexity and $\mathcal{O}(n^2)$ storage for n training data points during the implementation of Cholesky decomposition of K. To further improve the scalability of the proposed model, approximation and delayed updates of K should be introduced to alleviate the computational burden in the proposed approach.

C. Scalability

The scalability issue of the proposed battery SOC estimation and transfer learning method can be summarized into two aspects: 1) the inherent cubic computational burden growth with data size increasing by calculating the matrix inverse of the kernel matrix K and 2) the objective and the derivative factorization of the joint optimization for both GPR layers and neural network layers. To settle the aforementioned two issues, Gaussian kernel matrix structured sparse approximations and semi-stochastic asynchronous gradient descent are employed to enhance the scalability of the proposed structure.

1) Structured Sparse Approximations: Structured sparse approximation is a significant branch of scalable Gaussian processes. A structured kernel interpolation (SKI) procedure [40], an inducing point method, is introduced to reach rapid structure-exploiting inference. The SKI can be realized by the following procedure:

1) *Inducing points*, noted as $\mathbf{U} = \{\mathbf{u}_i\}_{i=1}^m$, are placed on equispaced grids as pseudo-inputs. In that way,

the subset-of-regressor (SoR) kernel approximation is reached as,

$$K_{X,X} \overset{\text{SoR}}{\approx} K_{X,U} K_{U,U}^{-1} K_{U,X}. \tag{10}$$

Here, the SoR approximate kernel gives rise to $\mathcal{O}(m^2 n + m^3)$ computations and $\mathcal{O}(mn + m^2)$ storage for GP inference and learning.

2) *Structured kernel interpolation*, such as local linear or cubic interpolation, is exploited to further reduce complexity. Linear interpolations for one-dimensional input are displayed as follows for simplicity:

$$\widetilde{k}(\mathbf{x_i}, \mathbf{u_j}) = \omega_i k(\mathbf{u_a}, \mathbf{u_j}) + (1 - \omega_i) k(\mathbf{u_j}, \mathbf{u_b}) \tag{11}$$

$$K_{X,U} \approx W K_{U,U} \tag{12}$$

where $\widetilde{k}(\cdot, \cdot)$ is the interpolated approximate kernel. And W is an $n \times m$ matrix of interpolation weights. Furthermore, W is extremely sparse. And $\mathbf{u_a}, \mathbf{u_b}$ are two inducing points most closely bound $\mathbf{x_i}$.
Substituting $K_{X,U}$ in eq.(12) into the SoR approximation for $K_{X,X}$ in eq.(10), this yields:

$$K_{X,X} \approx W K_{U,U} K_{U,U}^{-1} K_{U,U} W^\top = W K_{U,U} W^\top. \tag{13}$$

3) *Kernel structure exploiting*, represented by Toeplitz and Kronecker methods, utilizes the existing structure of the kernel matrix K to accelerate matrix calculation. For one-dimensional input, the kernel matrix is Toeplitz if it is generated from a stationary kernel, $k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x} - \mathbf{x}')$, with inputs $\mathbf{x}$ on a regularly spaced one-dimensional grid. Kronecker method is complementary to the Toeplitz method for multi-dimension inputs on a Cartesian grid for tensor product kernels. With the kernel structure exploiting, it makes inference and learning scale as $\mathcal{O}(n)$ and test predictions be $\mathcal{O}(1)$ while retaining model capability.

2) Semi-Stochastic Asynchronous Gradient Descent: It can be seen from (5) and (6) that the item $(K^{-1}\mathbf{y}\mathbf{y}^\top K^{-1} - K^{-1})$, can be pre-calculated on the full data so that the optimization of $\boldsymbol{\theta}$ and W will be benefited. The pre-calculation supports a stochastic mini-batch update for W and a deterministic full-batch update for $\boldsymbol{\theta}$ in a semi-stochastic style. Moreover, the stochastic updates of W are tiny enough, so that K does not change much between mini-batches. Within this context, multiple stochastic mini-batch gradient descents are performed to update recurrent network parameters before one full-batch conjugate gradient descents for GPR kernel matrix K in an asynchronous style. The procedure is given in Algorithm 1.

Algorithm 1 Semi-Stochastic Asynchronous Gradient Descent

Input: Data$-(X, \mathbf{y})$
1: Initialize $\boldsymbol{\theta}$ and w; compute initial K.
2: **repeat**
3: $\quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + update_{\boldsymbol{\theta}}(X, W, K)$.
4: $\quad$ **for all** mini-batches X_b in X **do**
5: $\quad\quad W \leftarrow W + update_W(X_b, W, K^{delayed})$
6: $\quad$ **end for**
7: $\quad$ Update the kernel matrix, K.
8: **until** Convergence
Output: Model $\mathbf{M}$ with optimal $\boldsymbol{\theta}^*$ and W^*.

D. Non-Parametric Fast Transfer Learning

Lithium-ion battery characteristics highly depend on the different ambient temperatures. The serious sensitivity to different temperatures poses major challenges on training a universal model to estimate the battery SOC at different temperatures. One promising strategy is to transfer the existing knowledge from the trained model to a new target of interest.

Transfer learning formulates a learning problem with domain and task, noted as $\mathcal{D}$ and $\mathcal{T}$, respectively. A domain $\mathcal{D}$ consists of two components, including a feature space $\mathcal{X}$ and a marginal probability distribution $P(X)$, where $X \in \mathcal{X}$. A task can be represented by $\mathcal{T} = \{\mathbf{y}, f(x)\}$, where $\mathbf{y}$ means the label space and $f(x)$ is the prediction function. In the battery SOC estimation problem, the domain mainly refers to the features extracted by neural networks and their distribution while the task is to estimate SOC at a specific temperature. And transfer learning is to learn a new target task $\mathcal{T}_t$ based on the target domain $\mathcal{D}_t$ while reusing or transferring information from the source domain $\mathcal{D}_s$ and the source task $\mathcal{T}_s$. The previously source task have the potential to significantly improve the training efficiency and model accuracy in target tasks. From the SOC estimation perspective, training a target model at a new temperature based on a previously trained source model at a different temperature is what needs to be done.

Transfer learning is a promising research topic in the machine learning field, which focuses on taking advantage of related but distinct knowledge from the previous tasks. Fine-tuning strategy is popular and easy to implement by freezing a portion of the parameters of a pre-trained model and retraining the remainder portion of the parameters. However, the parametric training is expensive owing to the large number of parameters to be trained. In addition, the tearing apart of a whole parametric model can potentially damage the connection between the frozen and retrained parameters resulting in a failure to transfer. Instead, a novel non-parametric fast transfer learning is designed to bypass burdensome parametric training while retaining satisfactory estimation accuracy. The CL-GPT freezes the whole parametric neural network part as a whole and updates the GPR layers with the target domain. In that way, the connection between parameters in the neural networks is preserved and able to execute a complete utility. The non-parametric advantages are mainly reflected in the swift fine-tuning of GPR layers. The GPR layers only involve several hyperparameters to be fine-tuned and the aforementioned structured sparse approximations can be exploited to accelerate transfer learning.

Theoretically, the neural network can be interpreted as a recurrent kernel for the Gaussian process from the perspective of the probabilistic GPR framework. In this sense, the input of GPR are features mapped by the recurrent kernel to capture implicit information independent from external conditions like temperatures. Based on essential features extracted by the neural network, the GPR then adapts to different conditions and achieves effective transfer learning with little data in target domains.

The hyperparameters globally adjust the mapping from the feature distribution to the prediction distribution. From the transfer learning perspective, the feature spaces do represent the output of neural networks, i.e., CNN and GRU. The feature spaces of source and target domain present similar but discrepant distributions. It is an efficient way to adjust distribution mapping by optimizing hyperparameters. The GP scheme also takes feature space from target domain as training data, i.e., $\mathbf{x}, \mathbf{y}$ in eq.(2). The non-parametric scheme does not rely on a multitude of parameters but rather few hyperparameters and features in target domain, which alleviates computational burden. A fast and accurate transfer learning is then achieved by taking non-parametric advantages.

E. Detailed Implementation

To clarify detailed procedure of the proposed method for SOC estimation at fixed or different temperatures, the step-by-step implementation is described as Fig. 3. The complete procedure of the proposed method can be divided into three phases: data process (Fig. 3(a) and (b)), model training (Fig. 3(c) and (d)), and inference (Fig. 3(e) and (f)). First, voltage (V), current (I), and temperature (T) are extracted to establish datasets for training, transfer, and testing from battery driving cycles. Most driving cycle data are classified as source domain data, noted as $(X_s, \mathbf{y_s})$. The remaining few driving cycle data are classified as target domain data, noted as $(X_t, \mathbf{y_t})$. Then, a pre-trained model, noted as $\mathbf{M_s}$, is trained using the source domain data via the semi-stochastic asynchronous gradient decent. This pre-trained model can estimate SOC at the fixed temperature from where the source domain data generate. Fine-tuning from the pre-trained model $\mathbf{M_s}$, a transfer learning model, noted as $\mathbf{M_t}$, can estimate SOC at different target temperatures employing the proposed fast transfer learning strategy. In the fast transfer learning, the network parameters W^* is frozen and the GPR layers of the $\mathbf{M_s}$ are updated using $(X_t, \mathbf{y_t})$. An algorithm for a detailed implementation for scalable recurrent structure non-parametric fast transfer learning is given in Algorithm 2.

Algorithm 2 Scalable Recurrent Structure With Non-Parametric Fast Transfer Learning

---

Input: Source domain data$-(X_s, \mathbf{y_s})$, target domain data$-(X_t, \mathbf{y_t})$, test point in source domain $\mathbf{x_{*s}}$, and test point in target domain $\mathbf{x_{*t}}$.

1: Train $\mathbf{M_s}$ using Algorithm 1 on $(X_s, \mathbf{y_s})$.
2: Compute $\mathbf{y_{*s}} \leftarrow \mathbf{M_s}(\mathbf{x_{*s}})$.
3: Transfer learning $\mathbf{M_t}$ by freezing network parameters W^* and updating GPR layers in $\mathbf{M_s}$ using $(X_t, \mathbf{y_t})$.
4: Compute $\mathbf{y_{*t}} \leftarrow \mathbf{M_t}(\mathbf{x_{*t}})$.

Output: Estimated value $\mathbf{y_{*s}}$ and $\mathbf{y_{*t}}$, models $\mathbf{M_s}$ and $\mathbf{M_t}$.

III. Experiments

In this section, the datasets and implementation setup for experimental validation are described. To comprehensively evaluate the proposed method, it is compared with several advanced estimation models in multiple aspects, such as SOC estimation accuracy, transfer learning performance, and time consumption.

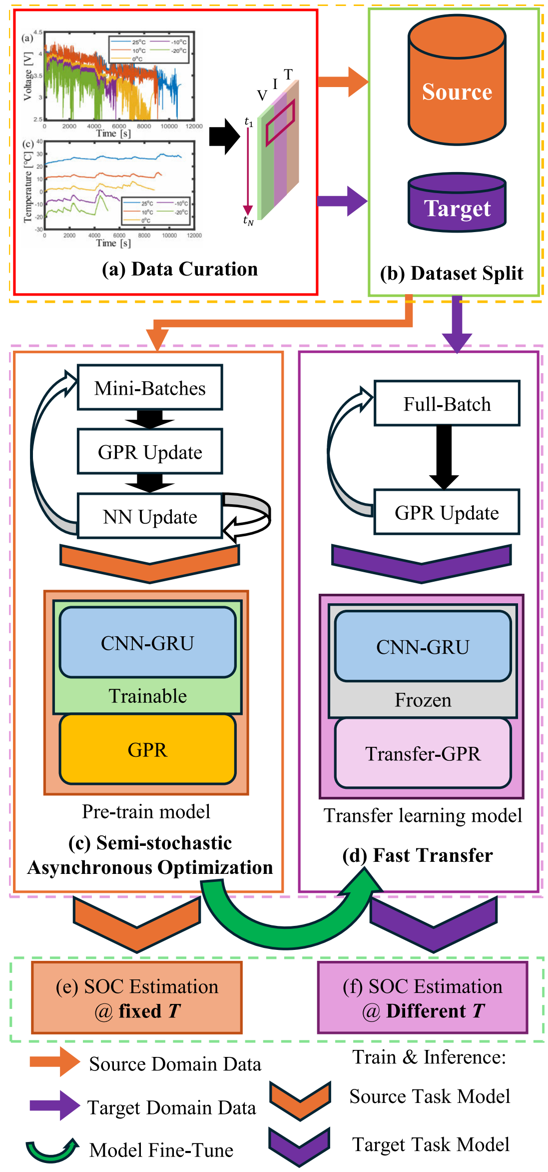

Fig. 3. Detailed implementation for SOC estimation at fixed and different temperatures of the proposed method.

A. Dataset Description

To extensively evaluate the SOC estimation performance of the proposed CL-GPT at constant and different temperatures, two datasets, i.e., Panasonic NCR 18650PF dataset [41], noted as Dataset 1, and LG 18650HG2 dataset [42], noted as Dataset 2, are employed for model training and transfer learning validation. Two manufacturers produce batteries in the two datasets with different chemistries and configurations, which ensures a variety of experiments.

A brand new 2.9Ah Panasonic 18650PF cell and a brand new 3Ah LG HG2 cell were tested in a controllable thermal chamber with a Digatron fire circuits universal battery tester

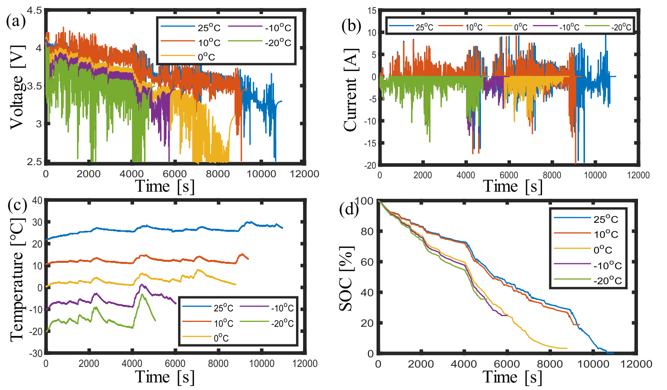

Fig. 4. Cycle1 drive cycle raw data at five ambient temperatures. (a) Voltage, (b) current, (c) battery surface temperature, and (d) SOC.

channel for Panasonic NCR 18650PF and LG 18650HG2 datasets, respectively. A series of tests were performed at five different ambient temperatures of 25 °C, 10 °C, 0 °C, −10 °C and −20 °C. For the Panasonic dataset, a range of drive cycles including Cycle1, Cycle2, Cycle3, Cycle4, US06, HWFET, UDDS, LA92, and Neural Network (NN) were performed at each temperature. Cycle 1-4 consist of random mix of US06, HWFET, UDDS, LA92 and NN drive cycles. NN drive cycle consists of a combination of portions of US06 and LA92 drive cycle and was designed to have some additional dynamics for training neural networks. For the LG dataset, four drive cycles, i.e., UDDS, HWFET, LA92, and US06, and eight random mixes of the four drive cycles, i.e., Mix 1-8, are performed at each temperature. For a new cell to be tested, it was cycled at 1C rate at 25 °C ten times to break it in. Then, a series of nine drive cycles were performed in a certain order. Batteries were charged after each test at 1C rate to 4.2V, 50mA cut off. The monitoring data, i.e., current, voltage, and battery surface temperature was measured at a 10 Hz sampling rate. To enhance computational efficiency, this study resampled the measured data to 1 Hz sampling rate. Fig. 4 shows the measured parameters and SOC values of Cycle1 at five ambient temperatures.

B. Implementation Setup

For fair comparisons, all the models are trained and tested on an Intel Core 13900K CPU platform using Keras 2.1.3 deep learning library. GPU is not involved in model training or testing, which makes it easy to migrate the proposed structures to onboard BMS devices.

The proposed CL-GPT structure is developed by concatenating CNN, GRU, and GPR layers in series as aforementioned. Its network specification and implementation details are summarized in Appendix. The CNN blocks are folded for simplification. The numbers in the specification column are alternatives used in different CNN blocks. The 2D CNN kernel and GRU hidden cell number are determined to be small but retain estimation accuracy.

C. Model Comparison

The proposed structure is compared with the commonly used advanced estimation model, such as CNN-LSTM [38] and CNN-GRU [43]. The CNN-LSTM and CNN-GRU are both tandem structures where a feature extraction module is followed by RNN components. They are capable of extracting spatial features and processing series data to estimate SOC. GPR is a classical method for prediction. But, solo GPR is not suitable for estimating SOC with monitoring data as it cannot capture long-term dependency and extract effective features. The solo GPR fails to map three-dimensional raw monitor data input into SOC estimate accurately. In order to compare effectively, the size of hidden layers and neurons in these neural networks are designed to be similar. Two datasets composed of Panasonic 18650PF and LG 18650HG2 batteries are both experimented to verify the generalization of the proposed methods.

TABLE I
HYPERPARAMETER SETTING FOR COMPARATIVE MODELS

Method	Seq Len	Hidden Size	Batch Size	Epoch	Optimizer(LR)	Dropout
CL-GPT	128	64	128	1000	Adam(1e-5)	0.1
CNN-LSTM	128	64	128	1000	Adam(1e-5)	0.1
CNN-GRU	128	64	128	1000	Adam(1e-5)	0.1

The experimental settings are displayed in Table I. Sequence length (Seq Len), Hidden Size, Batch Size, Optimizer, Learning Rate (LR), and Dropout Value are hyper-parameters to be tuned. The Seq Len determines the input length of GRU or LSTM. Longer Seq Len provides more comprehensive series information, enabling the recurrent layer to capture extended dependencies. Conversely, an excessively long Seq Len can reduce sensitivity to recent information. A high dimension of Hidden Size would improve the expression and ability to approximate nonlinearity but also bring a burden to training. The Batch Size of 128 is determined by iterating over powers of two to optimize the trade-off between gradient estimation stability and memory constraints. To ensure the structure is fully trained, Epoch is set big enough to make it converge for both tasks. For LR configuration, this article adapts 1e-5 for SOC estimation based on empirical validation. Regarding Dropout Value regularization, this study empirically set the rate to 0.1 after balancing model complexity and generalization, which effectively prevents overfitting.

For a fair comparison and demonstrate the superiority of the proposed CL-GPT method, all comparative models including CNN-LSTM, CNN-GRU, and the proposed method are thoroughly optimized and fully trained. The losses of models fully converge with adequate epochs, as illustrated in Fig. 5. The comparative models have similar neural networks with the same hyperparameters, e.g., CNN layers and recurrent hidden sizes, tuned by a comprehensive search.

1) SOC Estimation at Fixed Temperatures: To evaluate SOC estimation accuracy, the proposed method is compared with comparative models at different fixed ambient temperatures. At each fixed ambient temperature, Cycle 1-4, HWFET, UDDS, and NN drive cycles are selected as the training set. The testing dataset contains LA92 drive cycles for Dataset 1. For Dataset 2, the Mix 1-8, UDDS, and US06 driving cycles

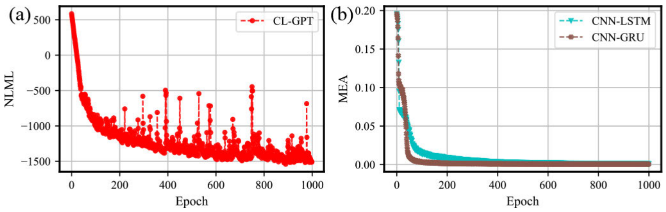

Fig. 5. Loss convergence curves. (a) NLML loss of CL-GPT, (b) MEA loss of CNN-LSTM and CNN-GRU.

are used as training set and the LA92 drive cycles are selected as testing dataset.

2) Fast Transfer SOC Estimation at Different Temperatures: To demonstrate the transfer learning performance of the proposed CL-GPT, training data at one temperature are used to pre-train a structure. Then, the pre-trained structure experiences transfer learning to estimate SOC at four different ambient temperatures. The transfer learning utilizes less training data at each corresponding target ambient temperature. Specifically, only Cycle 1 or Mix 1 drive cycle is selected as training data from each target ambient temperature domain for Dataset 1 or Dataset 2, respectively. The small amount of data for transfer learning fulfills the demand that less training data is available in target domain than those in source domain.

The proposed CL-GPT is transferred in the fashion mentioned in Section II-D. The whole neural networks are frozen and the GPR layer is adjusted to adapt to the target task. Instead, the compared models are transferred in a classic fine-tune way by freezing some layers and retraining the remainder layers. For CNN-LSTM and CNN-GRU models, the CNN part is frozen and the recurrent networks are retrained as CNN layers play a role of general feature extraction. And the shallow layers are frozen and the other part is adjusted for CNN model. As for LSTM and GRU models, the output dense layers are retrained using target domain data.

To highlight the fast transfer learning of the proposed structure, transfer time is a critical evaluation indicator. Transfer time is often ignored but is a significant evaluation dimension. Transfer learning is promising to apply in the real world for its flexibility and efficiency. Therefore, the transfer time should be emphasized and compared for different transfer learning models. In this work, transfer time is a concerned indicator and is compared with advanced transfer learning models at different ambient temperatures.

3) Scalability: The scalability techniques distinguish the proposed structure from traditional GPR-like methods. Scalability enables large amounts of data to be processed and contributes to deploying SOC estimation methods on BMS hardware devices. Training and inference time comparison is performed to demonstrate the scalability of the proposed structure. The time consumption in the training, inference, and transfer phases is analyzed to assess the feasibility for real-world applications.

D. Validation Metric

To compare the performance of CL-GPT with other SOC estimation and transfer methods, mean absolute error (MAE), root mean square error (RMSE), and maximum error (MAX) are used as the performance evaluation metrics:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\boldsymbol{y}_i - \tilde{\boldsymbol{y}}_i| \tag{14}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{y}_i - \tilde{\boldsymbol{y}}_i)^2} \tag{15}$$

$$\mathrm{MAX} = \max_i |\boldsymbol{y}_i - \tilde{\boldsymbol{y}}_i| \tag{16}$$

where N is the number of evaluated samples, $\boldsymbol{y}_i$ and $\tilde{\boldsymbol{y}}_i$ are the true SOC labels and the estimated SOC values for the i-th instance.

TABLE II

SOC Estimation Results at Different Ambient Temperatures (Dataset 1)

Method	Metrics	25°C	10°C	0°C	−10°C	−20°C	Avg.
CNN-LSTM	RMSE(%)	4.51	2.64	2.52	5.25	3.38	3.66
	MAE(%)	3.61	2.15	1.97	2.70	2.78	2.64
	MAX(%)	11.9	7.34	8.49	34.3	8.55	14.1
CNN-GRU	RMSE(%)	1.82	2.70	4.40	5.86	4.29	3.81
	MAE(%)	1.41	1.96	2.67	4.57	3.21	2.76
	MAX(%)	6.58	14.7	40.7	28.5	19.2	21.9
CL-GPT	RMSE(%)	**0.95**	**1.13**	**1.42**	**2.23**	**2.60**	**1.67**
	MAE(%)	**0.73**	**0.87**	**1.15**	**1.72**	**2.09**	**1.31**
	MAX(%)	**4.09**	**4.39**	**4.06**	**6.23**	**7.80**	**5.31**

Best results are highlighted in bold.

IV. Results and Analysis

A. SOC Estimation at Fixed Temperatures

The SOC estimation performance of the proposed deep recurrent structure CL-GPT is compared with other methods at fixed temperatures. The SOC estimation results of Dataset 1 are shown in Fig. 6. All comparative methods fit the SOC decreasing trends at different fixed temperatures generally. It is worth noting that the results of the proposed CL-GPT are the closest to the true SOC value from both global and zoomed figures. The proposed method holds the overall low errors, as displayed in Table II. Additionally, the proposed method provides faithful uncertainty CI uniquely. The uncertainty CI indicates the confidence of estimation results. The estimation results with narrower CIs are more reliable. It is noted that the CIs are narrower at higher ambient temperatures, i.e., 25°C, 10°C, and 0°C, which suggests that the estimation results are more reliable at these ambient temperatures. Conversely, the remaining comparative methods cannot provide a uncertainty indicator to reveal explanations about the given estimation results. Also, some comparative methods, e.g., CNN-GRU, estimate extremely inaccurate results in the early stages at some temperatures. The inaccuracy in the early stage indicates that these models cannot capture aging features in early life of batteries effectively.

To verify the generalization and robustness of the proposed methods, batteries in Dataset 2 are utilized for SOC estimation as they have different chemistry and configurations from Dataset 1. The SOC estimation results are shown in

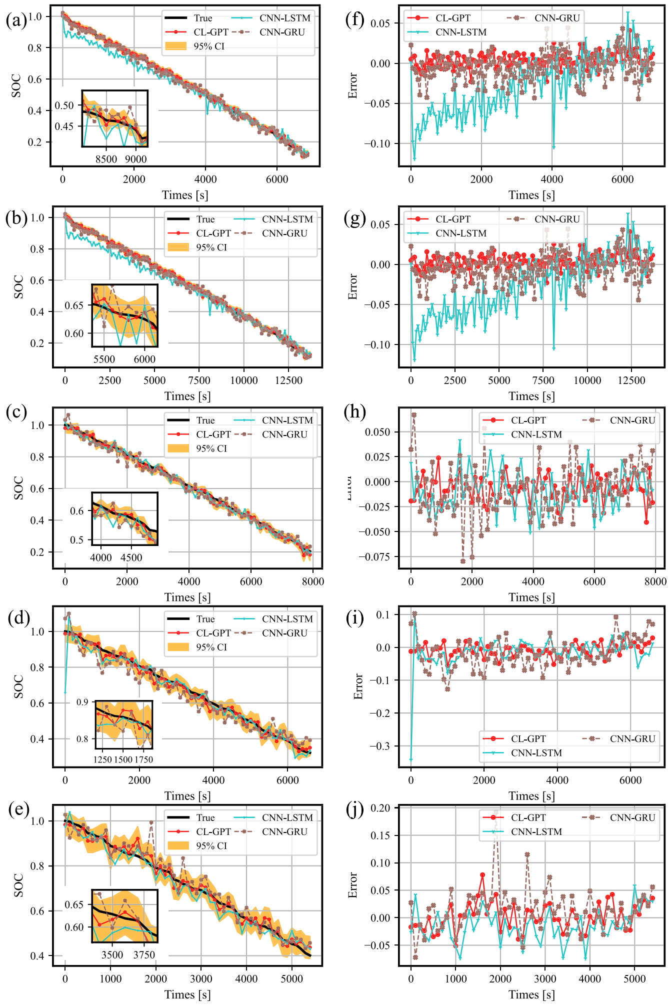

Fig. 6. The SOC estimation result comparison and errors at different fixed ambient temperatures (Dataset 1). (a)$\sim$(e) Result comparison at 25°C, 10°C, 0°C, −10°C, and −20°C, respectively. (f)$\sim$(g) Errors at 25°C, 10°C, 0°C, −10°C, and −20°C, respectively.

TABLE III

SOC Estimation Results at Different Ambient Temperatures (Dataset 2)

Method	Metrics	25°C	10°C	0°C	−10°C	−20°C	Avg.
CNN-LSTM	RMSE(%)	1.12	1.80	1.63	1.99	1.72	1.65
	MAE(%)	0.93	1.44	1.14	1.57	1.37	1.29
	MAX(%)	2.69	3.93	4.31	4.43	4.48	3.97
CNN-GRU	RMSE(%)	1.55	1.77	2.04	2.03	2.10	1.90
	MAE(%)	1.18	1.20	1.52	1.61	1.72	1.45
	MAX(%)	7.31	5.13	7.90	4.89	6.50	6.35
CL-GPT	RMSE(%)	**0.84**	**1.30**	**0.81**	**0.99**	**1.12**	**1.01**
	MAE(%)	**0.67**	**1.07**	**0.65**	**0.79**	**0.79**	**0.80**
	MAX(%)	**2.37**	**3.10**	**2.52**	**2.16**	**2.05**	**2.44**

Best results are highlighted in bold.

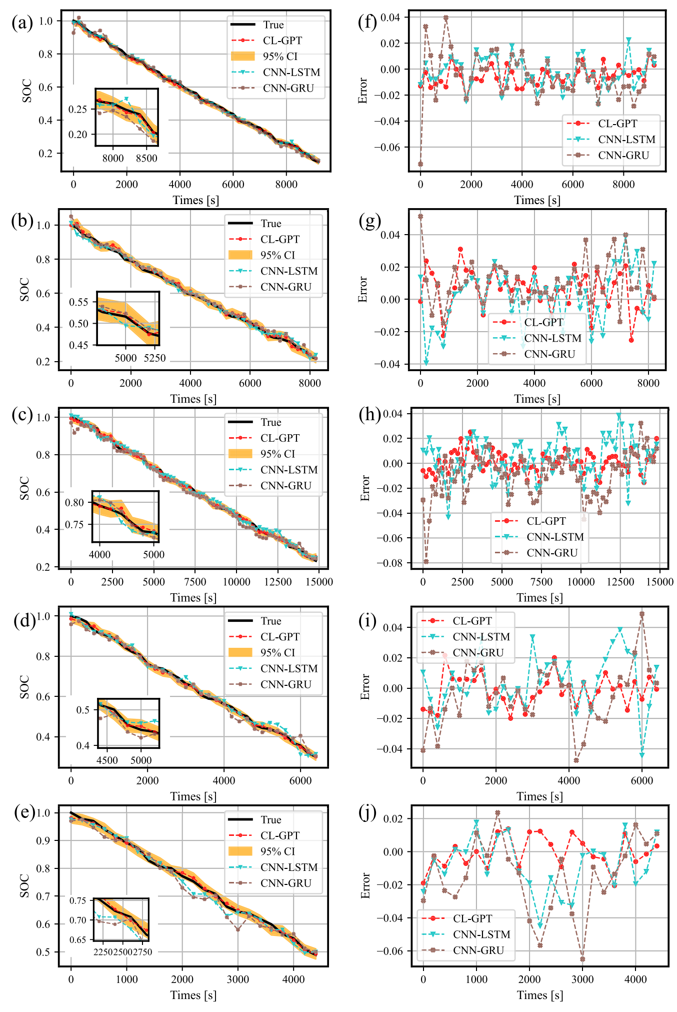

Fig. 7. The SOC estimation result comparison and errors at different fixed ambient temperatures (Dataset 2). (a)$\sim$(e) Result comparison at 25°C, 10°C, 0°C, −10°C, and −20°C, respectively. (f)$\sim$(g) Errors at 25°C, 10°C, 0°C, −10°C, and −20°C, respectively.

Fig. 7. It is worth noting that the proposed CL-GPT method estimated results fit the true SOC curves more smoothly than the comparative methods at all temperatures. The narrow CIs of CL-GPT also indicate faithful SOC estimations. Table III gives numeric SOC estimation results at different ambient temperatures. The proposed method holds the lowest average RMSE and MAE, i.e., 1.01% and 0.80%, respectively. The MAX metric evaluates the worst-case performance of an algorithm. We can infer from both Dataset 1 and Dataset 2 that the proposed Cl-GPT has the lowest MAX, indicating its high robustness. The error variations show that the CL-GPT has stable error fluctuations close to zero. The reduced error fluctuation amplitude indicates that it is stable and enables rapid error correction at a certain time step.

B. Transfer SOC Estimation at Different Temperatures

To validate the transfer performance of the proposed method under a little labeled data in target domain, comparative experiments are conducted on Dataset 1 from different source domains at different temperatures to target domain at the remaining temperatures.

As Fig. 8 shows, both RMSE and MAE under source pre-trained scenario grow as the difference in temperature between source domain and target domain increases. For instance, when taking 25°C as source domain, the RMSE and MAE are higher and higher as target temperatures are getting lower. It means that discrepancy in domain distribution grows when the difference in temperature increases. The results with

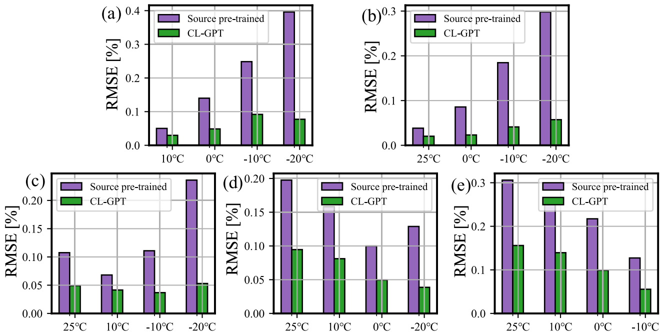

Fig. 8. Cross-domain SOC estimation results between direct and transfer learning from different ambient temperatures. (a) 25°C, (b) 10°C, (c) 0°C, (d) −10°C, and (e) −20°C.

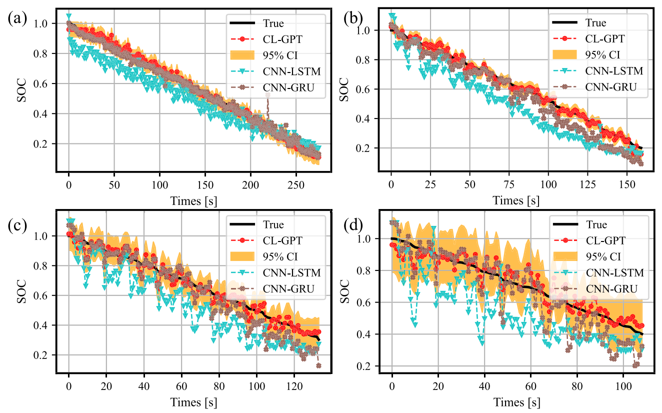

Fig. 9. Cross-domain SOC estimation result comparison from 10°C (source domain) to other temperatures(target domain) (Dataset 1). (a) 25°C, (b) 0°C, (c) −10°C, (d) −20°C.

TABLE IV

CROSS-DOMAIN SOC ESTIMATION RESULTS TRANSFERRING FROM 10°C(SOURCE DOMAIN) TO OTHER TEMPERATURES (TARGET DOMAINS) (DATASET 1)

Method	Metrics	10°C→25°C	10°C→0°C	10°C→−10°C	10°C→−20°C	Avg.
CNN-LSTM	RMSE(%)	9.55	14.1	18.3	24.5	16.7
	MAE(%)	8.39	13.1	16.0	20.7	14.5
	MAX(%)	18.6	21.7	31.2	48.7	30.1
CNN-GRU	RMSE(%)	2.64	7.12	7.89	11.1	7.18
	MAE(%)	1.92	5.99	6.17	8.55	5.65
	MAX(%)	22.7	13.9	22.0	28.9	21.8
CL-GPT	RMSE(%)	**2.03**	**2.31**	**4.11**	**5.73**	**3.54**
	MAE(%)	**1.65**	**1.81**	**3.24**	**4.69**	**2.85**
	MAX(%)	**7.67**	**6.26**	**11.4**	**14.0**	**9.83**

Best results are highlighted in bold.

TABLE V

CROSS-DOMAIN ESTIMATION RESULTS TRANSFERRING FROM ONE TEMPERATURE (SOURCE DOMAIN) TO OTHER TEMPERATURES (TARGET DOMAINS) (DATASET 1)

Method	Metrics	25°C →$\mathcal{R}$	10°C →$\mathcal{R}$	0°C →$\mathcal{R}$	−10°C →$\mathcal{R}$	−20°C →$\mathcal{R}$	Avg.
CNN-LSTM	RMSE(%)	24.6	16.6	28.0	19.6	20.2	21.8
	MAE(%)	21.9	14.5	24.3	17.1	16.4	18.8
CNN-GRU	RMSE(%)	8.48	7.18	6.85	11.6	**10.7**	10.3
	MAE(%)	6.86	5.66	5.61	9.11	**8.59**	7.16
Source pre-trained	RMSE(%)	13.7	15.4	24.6	18.1	12.5	16.7
	MAE(%)	11.8	13.9	22.8	15.5	9.87	14.8
CL-GPT	RMSE(%)	**6.19**	**3.54**	**4.51**	**6.60**	11.3	**6.43**
	MAE(%)	**4.67**	**2.85**	**3.50**	**5.34**	9.22	**5.12**

$\mathcal{R}$ denotes other four ambient temperatures.
Best results are highlighted in bold.

transfer learning satisfy the above pattern but hold much lower RMSE and MAE. This suggests that the proposed structure with fast transfer learning has a good performance on cross-domain estimation tasks.

The cross-domain transfer learning estimation results from 10°C, with no lack of generality, to other temperatures, are shown in Fig. 9. The compared methods fit the SOC consumption trends well. All compared methods except CNN-LSTM have appreciable accuracy under 25°C and 0°C target domain scenario. CNN-LSTM is not appreciated in this cross-domain task. Results of all methods, CNN-LSTM especially, deviate from the truth severally under temperatures below zero. It implies that battery characteristics change more linear and complicated during discharging in temperatures below zero. And low temperatures enlarge the discrepancy between source and target domain distribution.

Complete cross-domain SOC estimation results from 10°C are shown in Table. IV. The proposed CL-GPT has the best results with an average RMSE of 3.54%, MAE of 2.85%, and MAX of 9.83%, respectively. Moreover, CNN-GRU bearing more resemblance to the proposed structure outperforms the remaining methods with an average RMSE of 7.18%, MAE of 5.65%, respectively. It can infer that the CNN-GRU-like structures are more suitable for cross-domain tasks than LSTM-like structures for Dataset 1. However, both CNN-LSTM and CNN-GRU hold much higher MAX than CL-GPT. It indicates that they are less robust and cannot fast converge to desired trajectories, especially in early stages.

To validate the robustness and the generality of different source domains of the proposed structure, experiments under different temperatures are conducted and results are shown in Table. V. It is worth noting that the CL-GPT outperforms other baseline methods in all cases but −20°C → $\mathcal{R}$ with the best average RMSE of 6.43% and MAE of 5.12%, respectively. It demonstrates the stability and robustness of the proposed methods. CNN-GRU performs best on −20°C → $\mathcal{R}$ case with RMSE of 10.7% and MAE of 8.59%, respectively. Compared with source pre-trained models, which is pre-trained with the structure of GL-GPT on the source domain data only, the CL-GPT makes an average improvement of over 50% indicating its effectiveness on transfer learning.

To validate the adaptation of methods for different types of batteries. Dataset 2 is employed for transfer learning experiments. As shown in Fig. 10, both the non-parametric transfer CL-GPT and the fine-tune-based methods, i.e., CNN-LSTM and CNN-GRU, accomplish cross-domain SOC estimation task taking 10 °C as source domain. Numerically, the proposed CL-GPT achieves the best accuracy from Table VI. The CL-GPT has the lowest average RMSE and MAE for every

TABLE VI
CROSS-DOMAIN SOC ESTIMATION RESULTS TRANSFERRING FROM 10°C (SOURCE DOMAIN) TO OTHER TEMPERATURES (TARGET DOMAINS) (DATASET 2)

Method	Metrics	10°C→ 25°C	10°C→ 0°C	10°C→ −10°C	10°C→ −20°C	Avg.
CNN-LSTM	RMSE(%)	1.67	2.28	2.89	4.18	2.75
	MAE(%)	1.34	1.75	2.26	3.50	2.21
	MAX(%)	4.64	6.80	8.09	**7.81**	6.83
CNN-GRU	RMSE(%)	1.77	1.65	6.01	4.93	3.59
	MAE(%)	1.17	1.03	3.67	3.88	2.44
	MAX(%)	9.49	6.56	24.5	13.8	13.6
CL-GPT	RMSE(%)	**1.24**	**1.47**	**2.57**	**4.03**	**2.33**
	MAE(%)	**0.94**	**1.08**	**2.17**	**3.2**	**1.85**
	MAX(%)	**3.53**	**5.43**	**5.72**	9.83	**6.12**

Best results are highlighted in bold.

TABLE VII
CROSS-DOMAIN ESTIMATION RESULTS TRANSFERRING FROM ONE TEMPERATURE (SOURCE DOMAIN) TO OTHER TEMPERATURES (TARGET DOMAINS) (DATASET 2)

Method	Metrics	25°C $\to\mathcal{R}$	10°C $\to\mathcal{R}$	0°C $\to\mathcal{R}$	−10°C $\to\mathcal{R}$	−20°C $\to\mathcal{R}$	Avg.
CNN-LSTM	RMSE(%)	2.59	2.75	2.03	2.70	3.26	2.67
	MAE(%)	2.10	2.21	1.65	2.20	2.42	2.12
CNN-GRU	RMSE(%)	3.39	3.59	2.40	7.02	8.66	5.01
	MAE(%)	2.52	2.44	1.91	5.39	6.37	3.73
Source pre-trained	RMSE(%)	9.92	4.69	2.87	7.15	18.6	8.65
	MAE(%)	8.62	3.78	2.32	6.55	17.1	7.67
CL-GPT	RMSE(%)	**2.81**	**2.33**	**1.71**	**1.77**	**2.74**	**2.27**
	MAE(%)	**2.19**	**1.85**	**1.44**	**1.40**	**2.17**	**1.81**

$\mathcal{R}$ denotes other four ambient temperatures.
Best results are highlighted in bold.

source domain temperature from Table VII. The CNN-GRU model performs much worse than CNN-LSTM when source domain temperatures are below zero. But the CL-GPT, which has a similar structure to CNN-GRU, has excellent performance on transfer learning of temperatures below zero. The best accuracy of the proposed CL-GPT under various circumstances demonstrates its generalization, robustness, and effectiveness.

C. Validation of Scalability

Scalability is the distinctive advantage of the proposed scalable deep recurrent structure. Fast training time for SOC estimation under fixed temperatures and transfer learning to different temperatures are crucial to the practical application. Inference time indicates the efficiency when online execution. Time consumption is a vital metric of scalability and computation complexity. The consumption time of training, transfer learning, and online inference time are illustrated in Table. VIII. As for training time, the proposed CL-GPT holds the least training time consumption with 939 seconds in all cases. CNN-GRU needs relatively more time than other methods though better accuracy than the other remaining methods.

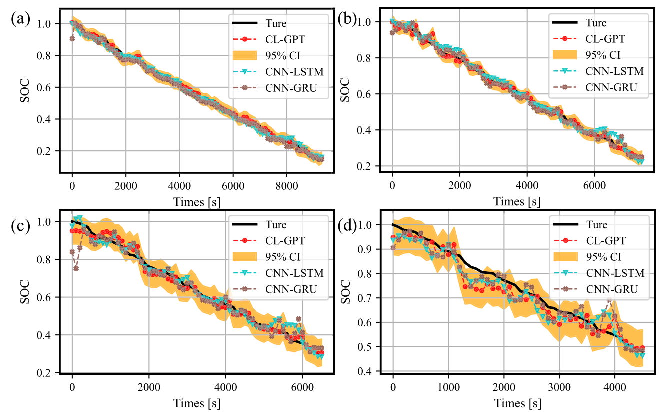

Fig. 10. Cross-domain SOC estimation result comparison from 10°C (source domain) to other temperatures(target domain) (Dataset 2). (a) 25°C, (b) 0°C, (c) −10°C, (d) −20°C.

TABLE VIII
TIME CONSUMPTION (S) OF TRAINING, TRANSFER AND INFERENCE AT DIFFERENT TEMPERATURES

Method	Type	25°C	10°C	0°C	−10°C	−20°C	Avg.
CNN-LSTM	Training	1771	1125	844	741	548	1006
	Transfer	13.5	14.3	14.7	15.7	16.2	14.9
	Inference	0.072	**0.063**	0.061	0.051	**0.036**	0.057
CNN-GRU	Training	2253	1939	1277	1057	930	1491
	Transfer	23.3	14.6	34.8	29.8	41.7	28.8
	Inference	**0.068**	**0.063**	**0.057**	**0.050**	**0.036**	**0.055**
CL-GPT	Training	**939**	**786**	**602**	**416**	**497**	**648**
	Transfer	**0.185**	**0.210**	**0.245**	**0.223**	**0.225**	**0.218**
	Inference	0.079	0.073	0.074	0.061	0.045	0.066

Best results are highlighted in bold.

The proposed scalable deep recurrent structure takes significant advantages considering transfer time. Table. VIII illustrates time consumption specifically. The proposed CL-GPT is two orders of magnitude faster than the ordinary fine-tuning strategy methods with an average transfer time of 0.218 seconds. It demonstrates the state-of-the-art performance on efficiency of the proposed method on fast transfer learning.

In practical applications, the inference time measures the computational time of a well-trained model for an online application. The inference time of CL-GPT is a little longer than the comparative methods. This is because an additional GPR layer is embedded in neural networks. However, a little longer inference time is acceptable in real-world applications since the inference time of these methods is all similar and as small as tens of milliseconds.

All comparative methods have the same settings on computational resources and execute training, transfer, and inference on the same dataset with the same data size for a fair comparison. The superior performance of the proposed method on accuracy and efficiency verifies that CL-GPT has faster convergence rates and better resource utilization rates since it works on the same data and provides more accurate and more rapid SOC estimation.

In addition to time comparison on specific devices, the computation complexity of the proposed method is evaluated by the disk space occupation and floating-point operations

TABLE IX
SPECIFICATION AND IMPLEMENTATION DETAILS OF THE PROPOSED STRUCTURE

Block	Type	Specification
Input Block	Input	batch size: 128, seq len: 128, feature num: 3
CNN Block1	Conv2D	out_channels: 16, kernel_size: 4×1
	Conv2D	out_channels: 32, kernel_size: 8×1
	Dense	out_features: 16
CNN Block2	Conv2D	out_channels: 64, kernel_size: 8×1
	Conv2D	out_channels: 64, kernel_size: 8×1
	Dense	out_features: 8
CNN Block3	Conv2D	out_channels: 128, kernel_size: 16×2
	Conv2D	out_channels: 64, kernel_size: 8×1
	Dense	out_features: 16
CNN Block4	Conv2D	out_channels: 64, kernel_size: 4×1
	Conv2D	out_channels: 64, kernel_size: 4×1
	Dense	out_features: 16
	Dense	out_features: 1
GRU Block	GRU layer	hidden_size: 64, dropout: 0.1
	Dense	out_features: 1
GPR Block	GPR layer	inf: infGrid, mean: meanZero, lik: likGauss
Output Block	Output	output_shape: 1

(FLOPs), which reflect the demands of real-world devices. The FLOPs are calculated by Tensorflow built-in model profiling API. The proposed CL-GPT has a relatively low disk space cost 335k FLOPs and disk space of 2,239 KB compared with classic CNN-based neural networks such as AlexNet [44] and VGG16 [45] which takes hundreds of MBs of disk space indicating the feasibility of deploying the proposed method in real-world applications.

V. CONCLUSION

This article proposes a scalable deep recurrent structure called CL-GPT for SOC estimation with uncertainty at different temperatures with a novel fast transfer learning fashion. The CL-GPT consists of several CNN blocks, GRU, and GPR layers. The proposed structure realizes effective information extraction from both local and long-term sequential data and accurate SOC estimation with quantified uncertainty. To transfer knowledge learned previously, the CL-GPT goes through a pre-training stage in the source domain and a fast transfer stage in the target domain. To demonstrate the state-of-the-art performance on accuracy and fast transfer learning, CL-GPT is compared with existing advanced methods such as CNN-LSTM, and CNN-GRU on the accuracy and time consumption dimensions. Abundant comparative experiment results validate that the CL-GPT outperforms these methods in estimation accuracy, efficiency, robustness, and predictive uncertainties. A limitation of the proposed method is that it does not consider battery aging effects on SOC estimation, given the inherent coupling between SOC and capacity. In further research, transfer learning between different aging effects will be worth investigating. It is worthwhile utilizing the scalable deep recurrent structure for other research interests such as state of health and SOC joint estimation.

APPENDIX

Specification and implementation details of the proposed structure are displayed for reproducibility and readability in Table IX.

REFERENCES

[1] Q. Yu, Y. Huang, A. Tang, C. Wang, and W. Shen, "OCV-SOC-Temperature relationship construction and state of charge estimation for a series–parallel lithium-ion battery pack," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 6, pp. 6362–6371, Jun. 2023.

[2] J. Tian et al., "Feature fusion-based inconsistency evaluation for battery pack: Improved Gaussian mixture model," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 446–458, Jan. 2023.

[3] R. Guo, C. Hu, and W. Shen, "An adaptive approach for battery state of charge and state of power co-estimation with a fractional-order multi-model system considering temperatures," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15131–15145, Dec. 2023.

[4] F. Ringbeck, M. Garbade, and D. U. Sauer, "Uncertainty-aware state estimation for electrochemical model-based fast charging control of lithium-ion batteries," *J. Power Sources*, vol. 470, Sep. 2020, Art. no. 228221.

[5] Q. Ouyang, Z. Wang, K. Liu, G. Xu, and Y. Li, "Optimal charging control for lithium-ion battery packs: A distributed average tracking approach," *IEEE Trans. Ind. Informat.*, vol. 16, no. 5, pp. 3430–3438, May 2020.

[6] Z. Xia and J. A. A. Qahouq, "State-of-charge balancing of lithium-ion batteries with state-of-health awareness capability," *IEEE Trans. Ind. Appl.*, vol. 57, no. 1, pp. 673–684, Jan. 2021.

[7] Q. Ouyang, W. Han, C. Zou, G. Xu, and Z. Wang, "Cell balancing control for lithium-ion battery packs: A hierarchical optimal approach," *IEEE Trans. Ind. Informat.*, vol. 16, no. 8, pp. 5065–5075, Aug. 2020.

[8] K. Liu, Y. Shang, Q. Ouyang, and W. D. Widanage, "A data-driven approach with uncertainty quantification for predicting future capacities and remaining useful life of lithium-ion battery," *IEEE Trans. Ind. Electron.*, vol. 68, no. 4, pp. 3170–3180, Apr. 2021.

[9] W. Zhu et al., "Uncertainty quantification of proton-exchange-membrane fuel cells degradation prediction based on Bayesian-gated recurrent unit," *eTransportation*, vol. 16, Apr. 2023, Art. no. 100230.

[10] J. Lu et al., "Battery degradation prediction against uncertain future conditions with recurrent neural network enabled deep learning," *Energy Storage Mater.*, vol. 50, pp. 139–151, Sep. 2022.

[11] L. Zheng, L. Zhang, J. Zhu, G. Wang, and J. Jiang, "Co-estimation of state-of-charge, capacity and resistance for lithium-ion batteries based on a high-fidelity electrochemical model," *Appl. Energy*, vol. 180, pp. 424–434, Oct. 2016.

[12] G. Dong, F. Yang, K.-L. Tsui, and C. Zou, "Active balancing of lithium-ion batteries using graph theory and A-star search algorithm," *IEEE Trans. Ind. Informat.*, vol. 17, no. 4, pp. 2587–2599, Apr. 2021.

[13] S. Marelli and M. Corno, "Model-based estimation of lithium concentrations and temperature in batteries using soft-constrained dual unscented Kalman filtering," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 2, pp. 926–933, Mar. 2021.

[14] Q. Ouyang, J. Chen, and J. Zheng, "State-of-charge observer design for batteries with online model parameter identification: A robust approach," *IEEE Trans. Power Electron.*, vol. 35, no. 6, pp. 5820–5831, Jun. 2020.

[15] K. D. Rao, A. H. Chander, and S. Ghosh, "Robust observer design for mitigating the impact of unknown disturbances on state of charge estimation of lithium iron phosphate batteries using fractional calculus," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3218–3231, Apr. 2021.

[16] Z. He, Z. Yang, X. Cui, and E. Li, "A method of state-of-charge estimation for EV power lithium-ion battery using a novel adaptive extended Kalman filter," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 14618–14630, Dec. 2020.

[17] X. Wu et al., "Online adaptive model identification and state of charge estimation for vehicle-level battery packs," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 596–607, Jan. 2024.

[18] G. Sethia, S. K. Nayak, and S. Majhi, "An approach to estimate lithium-ion battery state of charge based on adaptive Lyapunov super twisting observer," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 3, pp. 1319–1329, Mar. 2021.

[19] H. Ruan, B. Sun, W. Zhang, X. Su, and X. He, "Quantitative analysis of performance decrease and fast-charging limitation for lithium-ion batteries at low temperature based on the electrochemical model," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 1, pp. 640–650, Jan. 2021.

[20] M.-F. Ng, J. Zhao, Q. Yan, G. J. Conduit, and Z. W. Seh, "Predicting the state of charge and health of batteries using data-driven machine learning," *Nature Mach. Intell.*, vol. 2, no. 3, pp. 161–170, Mar. 2020.

[21] M. Aykol et al., "Perspective—Combining physics and machine learning to predict battery lifetime," *J. Electrochem. Soc.*, vol. 168, no. 3, Mar. 2021, Art. no. 030525.

[22] L. Xuan, L. Qian, J. Chen, X. Bai, and B. Wu, "State-of-charge prediction of battery management system based on principal component analysis and improved support vector machine for regression," *IEEE Access*, vol. 8, pp. 164693–164704, 2020.

[23] T. Berghout, M. Benbouzid, Y. Amirat, and G. Yao, "Lithium-ion battery state of health prediction with a robust collaborative augmented hidden layer feedforward neural network approach," *IEEE Trans. Transport. Electrific.*, vol. 9, no. 3, pp. 4492–4502, Mar. 2023.

[24] Z. Deng, X. Hu, X. Lin, Y. Che, L. Xu, and W. Guo, "Data-driven state of charge estimation for lithium-ion battery packs based on Gaussian process regression," *Energy*, vol. 205, Aug. 2020, Art. no. 118000.

[25] D. N. T. How, M. A. Hannan, M. S. H. Lipu, K. S. M. Sahari, P. J. Ker, and K. M. Muttaqi, "State-of-charge estimation of Li-ion battery in electric vehicles: A deep neural network approach," *IEEE Trans. Ind. Appl.*, vol. 56, no. 5, pp. 5565–5574, Sep. 2020.

[26] Z. Fei, Z. Zhang, F. Yang, and K.-L. Tsui, "Deep learning powered rapid lifetime classification of lithium-ion batteries," *eTransportation*, vol. 18, Oct. 2023, Art. no. 100286.

[27] M. A. Hannan, D. N. T. How, M. B. Mansor, M. S. H. Lipu, P. J. Ker, and K. M. Muttaqi, "State-of-charge estimation of Li-ion battery using gated recurrent unit with one-cycle learning rate policy," *IEEE Trans. Ind. Appl.*, vol. 57, no. 3, pp. 2964–2971, May 2021.

[28] Y. Tan and G. Zhao, "Transfer learning with long short-term memory network for state-of-health prediction of lithium-ion batteries," *IEEE Trans. Ind. Electron.*, vol. 67, no. 10, pp. 8723–8731, Oct. 2020.

[29] M. A. Hannan et al., "SOC estimation of Li-ion batteries with learning rate-optimized deep fully convolutional network," *IEEE Trans. Power Electron.*, vol. 36, no. 7, pp. 7349–7353, Jul. 2021.

[30] K.-H. Kim, K.-H. Oh, H.-S. Ahn, and H.-D. Choi, "Time–frequency domain deep convolutional neural network for Li-ion battery SoC estimation," *IEEE Trans. Power Electron.*, vol. 39, no. 1, pp. 125–134, Jan. 2024.

[31] Y. Che et al., "Battery states monitoring for electric vehicles based on transferred multi-task learning," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 10037–10047, Aug. 2023.

[32] M. Al-Shedivat, A. G. Wilson, Y. Saatchi, Z. Hu, and E. P. Xing, "Learning scalable deep kernels with recurrent structure," *J. Mach. Learn. Res.*, vol. 18, no. 1, pp. 2850–2886, Jan. 2017.

[33] G. O. Sahinoglu, M. Pajovic, Z. Sahinoglu, Y. Wang, P. V. Orlik, and T. Wada, "Battery state-of-charge estimation based on regular/recurrent Gaussian process regression," *IEEE Trans. Ind. Electron.*, vol. 65, no. 5, pp. 4311–4321, May 2018.

[34] Z. Ni, B. Li, and Y. Yang, "Deep domain adaptation network for transfer learning of state of charge estimation among batteries," *J. Energy Storage*, vol. 61, May 2023, Art. no. 106812.

[35] J. Lu, R. Xiong, J. Tian, C. Wang, and F. Sun, "Deep learning to estimate lithium-ion battery state of health without additional degradation experiments," *Nature Commun.*, vol. 14, no. 1, p. 2760, May 2023.

[36] S. Kumayl Raza Moosavi et al., "Transductive transfer learning-assisted hybrid deep learning model for accurate state of charge estimation of Li-ion batteries in electric vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 10, pp. 14445–14459, Oct. 2024.

[37] K. Liu et al., "Transfer learning for battery smarter state estimation and ageing prognostics: Recent progress, challenges, and prospects," *Adv. Appl. Energy*, vol. 9, Feb. 2023, Art. no. 100117.

[38] L. Shen, J. Li, J. Liu, L. Zhu, and H. T. Shen, "Temperature adaptive transfer network for cross-domain state-of-charge estimation of Li-ion batteries," *IEEE Trans. Power Electron.*, vol. 38, no. 3, pp. 3857–3869, Mar. 2023.

[39] C. K. Williams and C. E. Rasmussen, *Gaussian Processes for Machine Learning*, vol. 2. Cambridge, MA, USA: MIT Press, 2006, ch. 5.

[40] A. G. Wilson and H. Nickisch, "Kernel interpolation for scalable structured Gaussian processes (KISS-GP)," in *Proc. 32nd Int. Conf. Int. Conf. Mach. Learn.*, vol. 37, Jul. 2015, pp. 1775–1784.

[41] E. Chemali, P. J. Kollmeyer, M. Preindl, R. Ahmed, and A. Emadi, "Long short-term memory networks for accurate state-of-charge estimation of Li-ion batteries," *IEEE Trans. Ind. Electron.*, vol. 65, no. 8, pp. 6730–6739, Aug. 2018.

[42] C. Vidal, P. Kollmeyer, E. Chemali, and A. Emadi, "Li-ion battery state of charge estimation using long short-term memory recurrent neural network with transfer learning," in *Proc. IEEE Transport. Electrific. Conf. Expo (ITEC)*, Jun. 2019, pp. 1–6.

[43] F. Wu, S. Wang, D. Liu, W. Cao, C. Fernandez, and Q. Huang, "An improved convolutional neural network-bidirectional gated recurrent unit algorithm for robust state of charge and state of energy estimation of new energy vehicles of lithium-ion batteries," *J. Energy Storage*, vol. 82, Mar. 2024, Art. no. 110574.

[44] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. 25th Int. Conf. Neural Inf. Process. Syst.*, May 2017, pp. 1097–1105.

[45] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. (ICLR)*, San Diego, CA, USA, May 2015, doi: 10.48550/arXiv.1409.1556.

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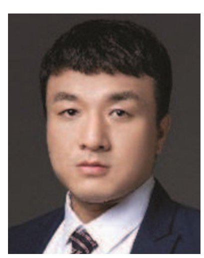

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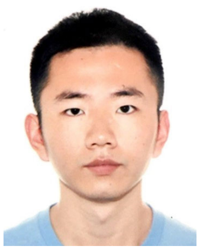

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